

**AMENDMENT NO. 3 SEPTEMBER 2013**  
**TO**  
**IS 3508 : 1966 METHODS OF SAMPLING AND TEST FOR**  
**GHEE ( BUTTERFAT )**

*(First Revision)*

(Page 26, clause **12.4.3**) — Substitute the following for the existing clause:

**‘12.4.3 Dilute Sulphuric Acid** – Use 1N sulphuric acid ( $\text{H}_2\text{SO}_4$ ) prepared by adding 27 ml of concentrated  $\text{H}_2\text{SO}_4$  (specific gravity 1.84) in 1 litre distilled water. Adjust so that 40 ml of 1 N  $\text{H}_2\text{SO}_4$  neutralize 2 ml of the 50 percent sodium hydroxide solution.’

(Page 56, clause **22.1**) — Substitute the following for the existing clause:

**‘22.1 Iodometric Method**

**22.1.1 Principle**

The material in an acetic acid-chloroform medium, is treated with an aqueous solution of potassium iodide. The liberated iodine is titrated with standard sodium thiosulphate solution.

**22.1.2 Apparatus and Glassware**

**22.1.2.1 Pipette**, graduated, 1 ml capacity.

**22.1.2.2 Conical flask**, glass-stoppered, 250 ml capacity or Iodine flask.

**22.1.3 Reagents**

**22.1.3.1 Acetic acid-chloroform solution** — Mix three parts by volume of glacial acetic acid, with 2 parts by volume of chloroform.

**22.1.3.2 Potassium iodide solution** — Saturated. Prepare saturated solution of potassium iodide in recently boiled distilled water. Store in the dark.

**22.1.3.3 Sodium thiosulphate solution** — 0.1 N, accurately standardized.

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**22.1.3.4** *Sodium thiosulphate solution* — 0.01 N. This solution is prepared by diluting 100 ml of accurately standardized solution of 0.1 N sodium thiosulphate to 1 000 ml with freshly boiled and cooled distilled water.

**22.1.3.5** *Starch solution* — 1 percent, prepare fresh.

#### **22.1.4** *Procedure*

Weigh  $5.00 \pm 0.05$  g of sample of fat in a 250 ml glass stoppered conical flask and then add 30 ml of the acetic acid-chloroform solution. Swirl the flask until the sample is dissolved. Add 0.5 ml of saturated potassium iodide solution. Allow the solution to stand exactly for 1 min with occasional shaking and then add 30 ml of distilled water. Titrate with 0.1 N sodium thiosulphate solution with constant and vigorous shaking. Continue titration until the yellow colour almost disappears, Add 0.5 ml of starch solution and continue titration till the blue colour just disappears. If the titre value is less than 0.5 ml, repeat the determination using 0.01 N sodium thiosulphate solution. Conduct a blank determination of the reagents in the same way. The titration in blank determination should not exceed 0.1 ml of the 0.1 N sodium thiosulphate solution.

#### **22.1.5** *Calculation*

$$\text{Peroxide Value (milli eqvt peroxide / 1 000g fat/ oil)} = \frac{(S-B) \times N \times 1\,000}{g}$$

where

$S$  = volume of sodium thiosulphate solution used up by the sample, in ml;

$B$  = volume of the sodium thiosulphate solution used up in the blank determination, in ml;

$N$  = normality of the sodium thiosulphate solution; and

$g$  = weight of the sample, in g.'

(Page 58, clause **22.2**) — Substitute the following for the existing clause:

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### **‘22.2 Spectrophotometric Method for Determination of Peroxide Value in Fat**

#### **22.2.1 General**

This method describes the determination of the peroxide value of anhydrous milk fat/ghee/butter-oil having peroxide value less than 1.0.

#### **22.2.2 Principle**

Solution of a weighed amount of the sample in a mixture of chloroform and methanol and addition of some iron (II) chloride and ammonium thiocyanate. After a fixed reaction time, photometric determination of the amount of red iron (III) complex.

#### **22.2.3 Reagents**

**22.2.3.1 Chloroform-methanol solution** — Mix 70 parts of chloroform ( $\text{CHCl}_3$ ) and 30 parts anhydrous methanol ( $\text{CH}_3\text{OH}$ ) by volume.

**22.2.3.2 Ferrous chloride** — Prepare this solution in diffused, indirect light. Dissolve 0.4 g of barium chloride ( $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$ ) in 50 ml water. Dissolve 0.5 g iron (II) sulphate ( $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ ) in 50 ml water. Slowly pour barium chloride solution with constant stirring into iron (II) sulphate solution and add 2 ml of concentrated. HCl. Allow the precipitate of barium sulphate to settle or centrifuge the mixture until upper liquid layer is clear. Decant the clear solution in brown bottle. Do not store solution for more than 1 week.

**22.2.3.3 Ammonium thiocyanate solution** — Dissolve 30 g ammonium thiocyanate ( $\text{NH}_4\text{SCN}$ ) in water and make up the volume to 100 ml. If solution is not colourless, remove the colour by extracting the solution several times with small amount of *iso*-amyl alcohol (3-methylbutanol).

**22.2.3.4 0.2 N hydrochloric acid (HCl) solution** — Dissolve 1.96 ml of HCl (sp gravity 1.16) in 100 ml distill water or dilute 2 ml of about 10 N hydrochloric acid with water to 100 ml.

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**22.2.3.5** *Standard solution of iron (III) chloride* — Dissolve 0.5 g iron powder in 50 ml of concentrated HCL and add 2 ml of 30 percent (w/w) hydrogen peroxide solution. Remove excess of  $\text{H}_2\text{O}_2$  by boiling for 5 min over boiling water bath. Cool to room temperature and make up volume to 500 ml with distilled water.

**22.2.3.6** *Working standard solution* — Pipette 1 ml of this standard solution and further dilute it to 100 ml with chloroform-methanol solution. Strength of this solution is 10  $\mu\text{g}$  of iron per ml.

#### **22.2.4** *Apparatus/Glassware*

**22.2.4.1** *Analytical balance*, sensitivity 1 mg.

**22.2.4.2** *Burettes*, of 10 ml capacity, with scale divisions at each 0.02 ml.

**22.2.4.3** *Pipettes*, of 1 ml capacity, provided with graduation marks at each 0.05 ml.

**22.2.4.4** *Spectrophotometer* — for measuring at a wavelength of 500 nm, a light-path of 1.0 cm.

#### **22.2.5** *Procedure*

##### **22.2.5.1** *Preparation of standard curve*

Prepare standard curve of 5  $\mu\text{g}$  - 50  $\mu\text{g}$  of Fe by pipetting 0.5 ml - 5 ml in increments of 0.5 ml working standard solution respectively into 9.4 ml - 4.9 ml of mixture of chloroform and methanol. Add 0.05 ml of ammonium thiocyanate solution and 0.05 ml of 0.2N HCl through micro-pipette to each of these standard tubes and mix. Wait for 5 min. Measure extinctions against mixture of chloroform and methanol. Plot the extinctions of series against added amounts of iron (III) ion, expressed as  $\mu\text{g}$  Fe (III).

##### **22.2.5.2** *Preparation of sample*

Weigh 0.3 gm of sample in a test tube. Add 9.6 ml of chloroform-methanol mixture and dissolve fat in it. Add 0.05 ml of ammonium thiocyanate solution

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and mix. Measure the extinctions against chloroform-methanol mixture at 500 nm. Mark it as  $E_{\text{fat blank}}$ . Add 0.05 ml of ferrous chloride and mix. Wait for 5 min. Then measure the extinctions against mixture of chloroform-methanol mixture at 500 nm. Mark it as  $E_{\text{sample}}$ . Carry out blank test by mixing 9.6 ml of chloroform-methanol mixture with 0.05 ml of ammonium thiocyanate and 0.05 ml of ferrous chloride and taking extinctions against mixture of chloroform-methanol mixture at 500 nm. Mark it as  $E_{\text{reagent blank}}$ . Complete full estimation within 10 min.

#### 22.2.5.3 Expression of results

Difference of extinctions (a) =  $E_{\text{Sample}} - (E_{\text{fat blank}} + E_{\text{reagent blank}})$

Calculate net iron from the standard curve, in  $\mu\text{g}$ .

Peroxide value, as milligramme equivalent per kg = 
$$\frac{\text{Net iron (in } \mu\text{g) from curve}}{55.84 \times m}$$

where

$m$  = weight of sample taken for analysis.

#### 22.2.6 Precautions

- a) Exposure of the sample of anhydrous milk fat to light shall be avoided.
- b) Carry out the test in indirect light, subdued as much as is practicable.'